BUS 9430 · CAPSTONE 3 · BACKUP SLIDES FOR QUESTIONS

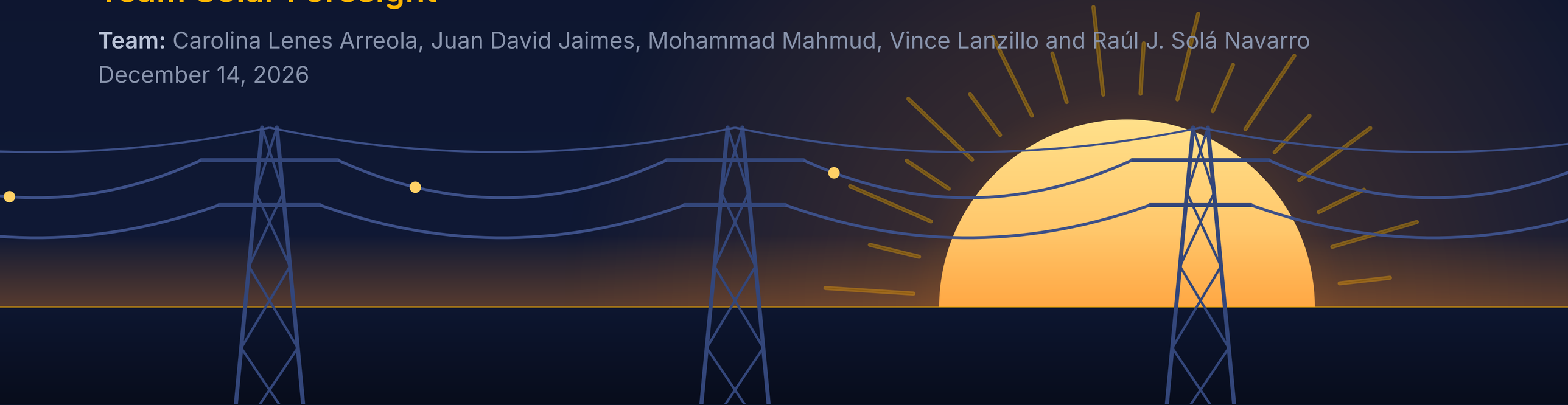
Tomorrow’s Sun, Today’s Plan

Detail behind the numbers, for the question period

Team Solar Foresight

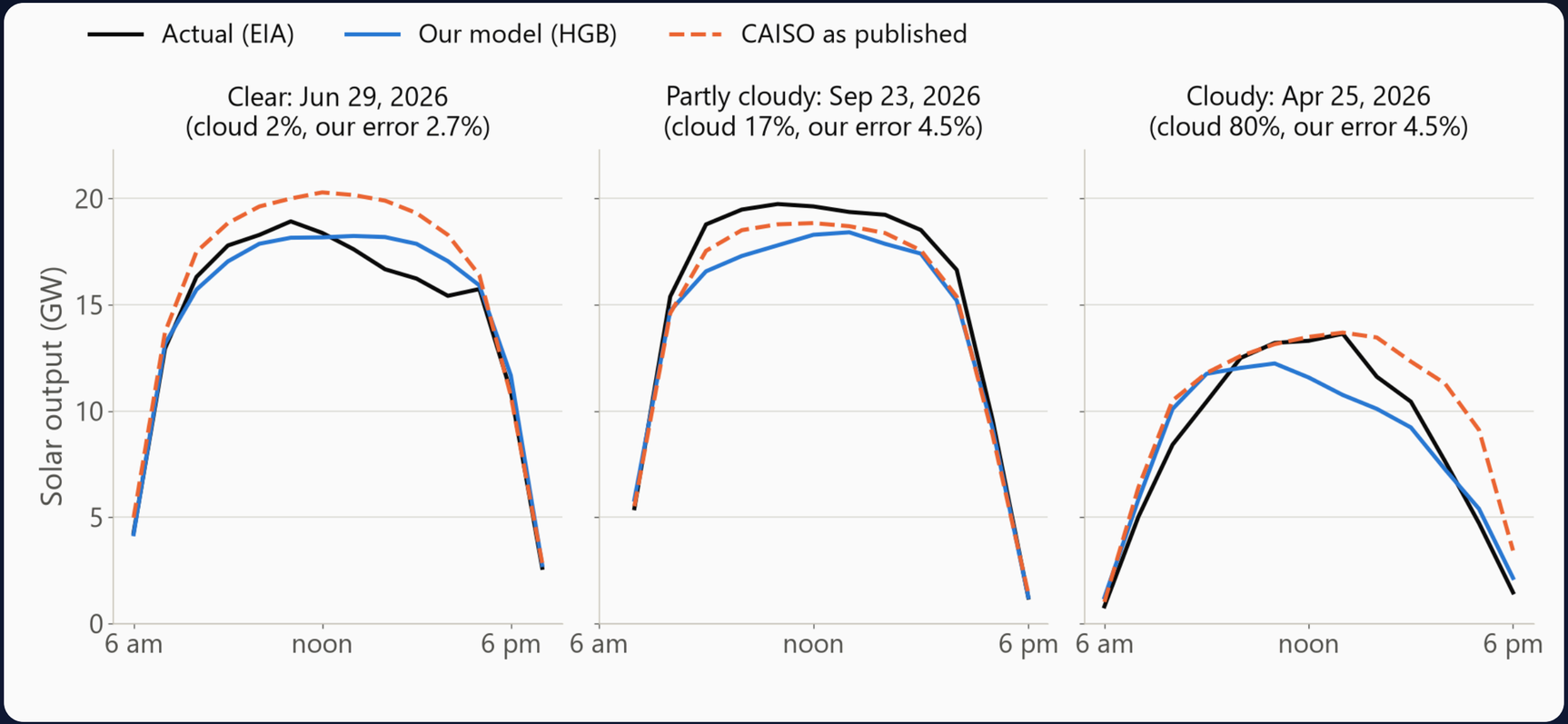
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December 14, 2026



BACKUP: WHAT A FORECAST DAY LOOKS LIKE

Three ordinary test days, forecast against actual

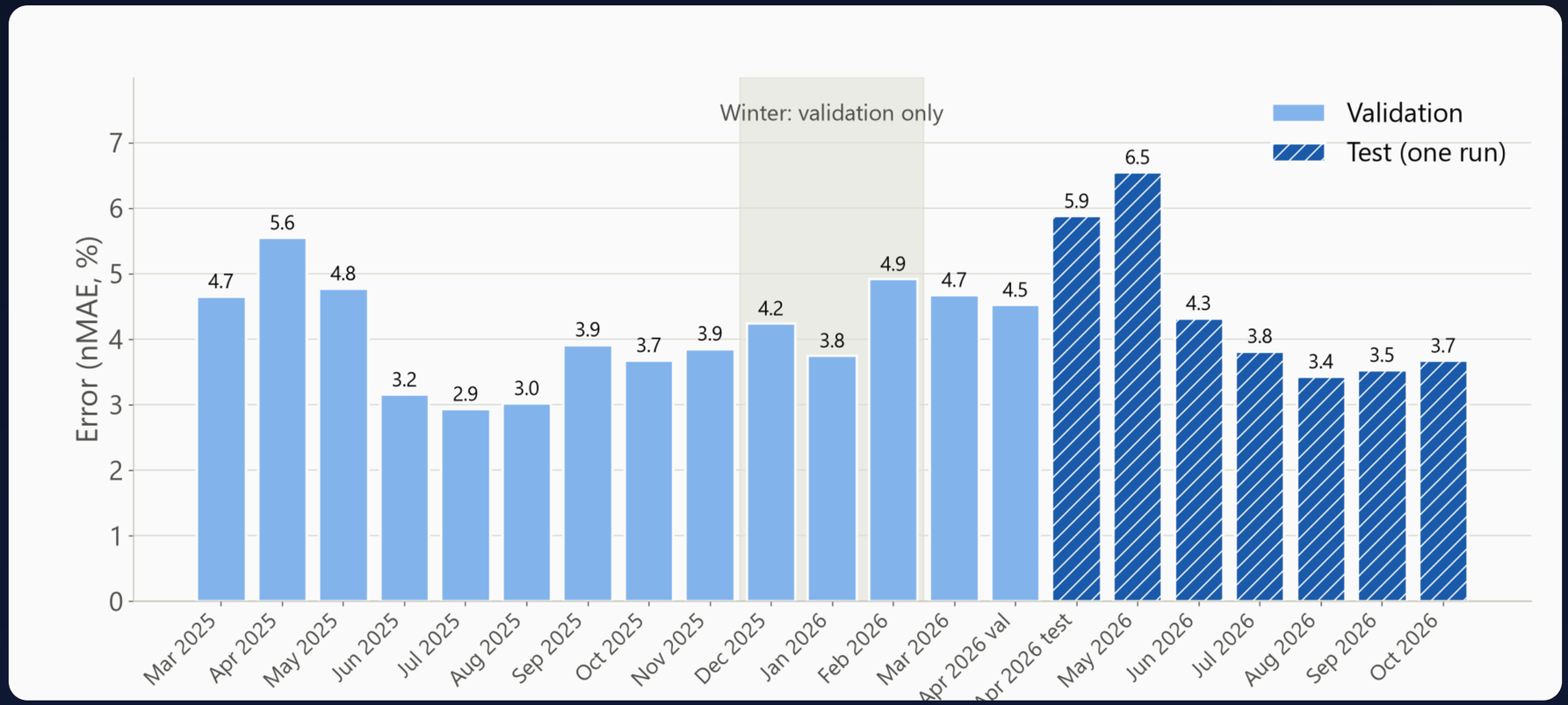


Each day is the **median-error day** in its cloud group, chosen by rule, not by hand. Our error that day: 2.7%, 4.5%, 4.5%. Solar output is in gigawatts (GW).

Median-error day among test days in each group of average day-2 cloud forecast (0 to 10, 10 to 60, above 60%). Pacific time. Actual: EIA hourly solar, CISO.

BACKUP: ERROR BY MONTH

Our error by month: the test run has no winter



Winter months (Dec 4.2%, Jan 3.8%, Feb 4.9%) appear only in the practice runs. **On the test, April and May 2026 were our worst months: 5.9% and 6.5%.**

Light bars: validation (practice runs on earlier time blocks). Hatched dark bars: the one test run. April 2026 is split at the test start (Apr 10). Validation spans Mar 2025 to Apr 9, 2026.

BACKUP: PRACTICE RUNS AGAINST THE TEST

Our error rose from 4.0% to 4.4%, mostly in spring

MEASURE	PRACTICE RUNS	TEST
Our error	4.02%	4.43%
CAISO as published	4.51%	5.85%
Same as 2 days ago	6.60%	7.27%
Our error in spring	4.95%	6.34%
Our average miss (forecast minus actual)	-186 MW	+494 MW

Why: April and May 2026 had 24.0% and 20.7% of solar turned down, against 13.2% and 6.6% a year earlier. The model learned from earlier, lower turn-down levels, so on the test it forecast too high.

Practice runs: 4,508 hours on 374 days. Test: 2,457 hours on 187 days. Source: reports/test_vs_validation.csv, FAQ Decision D31. 141 of 231 comparisons moved by more than 0.5 points.

BACKUP: CAISO ADJUSTED

Why the adjusted CAISO score is not a fair yardstick

WHAT WE CHECKED

Turn-downs match CAISO's own over-forecast

CAISO measures turned-down solar against its own forecast. In hours with large turn-downs, the two numbers add up to 104%, correlation 0.89.

WHAT IT MEANS

Adding it back removes most of CAISO's error

In the 777 heaviest hours, CAISO's error is 11.75% as published and 2.77% adjusted. So the adjusted score is low partly by construction.

So our headline comparison is CAISO as published, and we show adjusted as a second line. On the test, adjusted is 3.54%.

Checked on the 4,508 validation hours (FAQ Decision D27).

BACKUP: HOW THE MODEL WORKS

20 inputs, all known at 10 a.m.

WEATHER FORECAST, 2 DAYS BEFORE (10)

Day-2 forecasts of cloud cover, sunshine, temperature, humidity, wind and rain, plus a cloud index against a clear sky.

TIME AND SUN (7)

Hour of day, hour and day of year as smooth curves, sun height, clear-sky sunshine.

SIZE AND RECENT OUTPUT (3)

Installed capacity that month, output 2 days earlier, average output on the day 2 days earlier.

Left out on purpose: output from 1 day earlier, CAISO's forecast, and day-1 weather, because they arrive after the 10 a.m. bid close.

FAQ Decision D28. Two assumptions are not verified: the 2-day-old weather forecast is available by 10 a.m., and EIA's values from 2 days earlier are published by then.

BACKUP: THE 10 A.M. RULE

Respecting the 10 a.m. rule costs about 0.15 points

WITH THE RULE (WHAT A REAL BIDDER COULD DO)

4.02% error

2-day-old weather forecast and output from 2 days earlier.

WITHOUT IT (COMPARISON ONLY)

3.87% error

1-day-old weather forecast. Better, but available too late to use in a bid.

The rule is a small price, and it keeps the score honest.

Practice runs, 4,508 hours (FAQ Decisions D28, D30).

BACKUP: BACKUP POWER IN DETAIL

How much backup, for how many hours

FORECAST	90 OF 100 HOURS	95 OF 100 HOURS	99 OF 100 HOURS
Our model	2,650 MW	3,630 MW	5,740 MW
Same as 2 days ago	3,230 MW	4,790 MW	8,470 MW
CAISO as published	4,260 MW	5,890 MW	8,290 MW

Our model by season at 95 of 100 hours: spring 4,150 MW, summer 3,250 MW, fall 2,840 MW. The 99 figure rests on about 25 hours.

MW for today's fleet: share of capacity times EIA 860M nameplate Sep 2026, all CISO, 26,034.5 MW. Hours when solar was turned down count as shortfalls, so CAISO's figures are overstated most (spring alone: 7,770 MW).

BACKUP: WHAT THE DOLLAR FIGURE MEANS

About $120 million a year: a size, not a savings promise

STEP 1

2.83 points

Error gap between the same-as-2-days-ago guess (7.27%) and our model (4.43%).

STEP 2

3.0 TWh a year

One point of error is about 1.06 TWh a year (Capstone 2 numbers). 2.83 times 1.06.

STEP 3

About $122 million

At $41 per MWh, CAISO's 2024 average day-ahead price. 2.83 times $43 million a point.

It sizes the problem. It does not say how much money a grid operator would save, and we do not compute a dollar figure against CAISO's own forecast.

Source: Capstone 2 numbers file (deck/capstone2/CAPSTONE2_NUMBERS.md) and reports/reserve_test.csv.

BACKUP: KEEPING THE TEST HONEST

We saved the rules first, then ran the test once

BEFORE

Model and rules frozen

Model settings, 20 inputs and five questions with pass rules, committed before the run (commit c242a1c).

DURING

One run

The script only reads test rows with an explicit flag, and it is meant to run once.

AFTER

Report what came out

Four questions supported, one partly. The miss of 0.005 points stays a miss.

Files: reports/final_model_spec.md, reports/hypotheses.md, src/modeling/final_test.py (team repository).

BACKUP: LIMITS AND OPEN ITEMS

What we have not checked or cannot show

- **No winter in the test.** Winter was checked only in practice runs.
- **2.6 years of history.** Each month appears 2 or 3 times.
- **Two timing assumptions.** The 2-day-old weather forecast is available by 10 a.m., and EIA's values from 2 days earlier are published by then. Neither is verified.
- **One capacity source.** Errors are shares of EIA 860M nameplate, all CISO, and we name it next to every MW figure.
- **Turned-down solar is a rough series.** It is above zero in 94% of hours, probably because CAISO defines it against its own forecast.
- **Forecast high on test.** Bias +494 MW, cause only partly understood.

FAQ open items and Decision D31 (team repository).

BACKUP: SOURCES

Sources and where to find everything

- **Solar output and capacity:** U.S. Energy Information Administration (hourly grid monitor, Form 860M).
- **Benchmark forecast and turn-downs:** CAISO OASIS and CAISO curtailment reports.
- **Weather forecasts:** Weather data by Open-Meteo.com (CC BY 4.0).
- **Dollar price:** CAISO 2024 average day-ahead price, $41 per MWh.
- **Code, numbers and decisions:** team repository, README "Start here", deck/capstone3/CAPSTONE3_NUMBERS.md and docs/DATA_FAQ.md (Decisions D27 to D31).